

ADMINISTRATION RESEAUX

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PLAN

- Rapelle sur GNS3 configuration
- Rapelle sur le VLAN (trunk, access, ip adresse switch, telnet switch and ssh switch), VTP, DTP
- Introduction to STP
- Introduction to access list standard and extended
- Automatisation de tache avec python
 - telnetlib**, paramiko**
(netmiko,NAPALM,Ansible,Nornir)

TRAVAIL PRATIQUE (simulateur et physique)

Logicielle de supervision

Nagios Core

Zabbix

Ganglia

Observium

Cacti

Icinga 2

Automatisation de tache

Netmiko

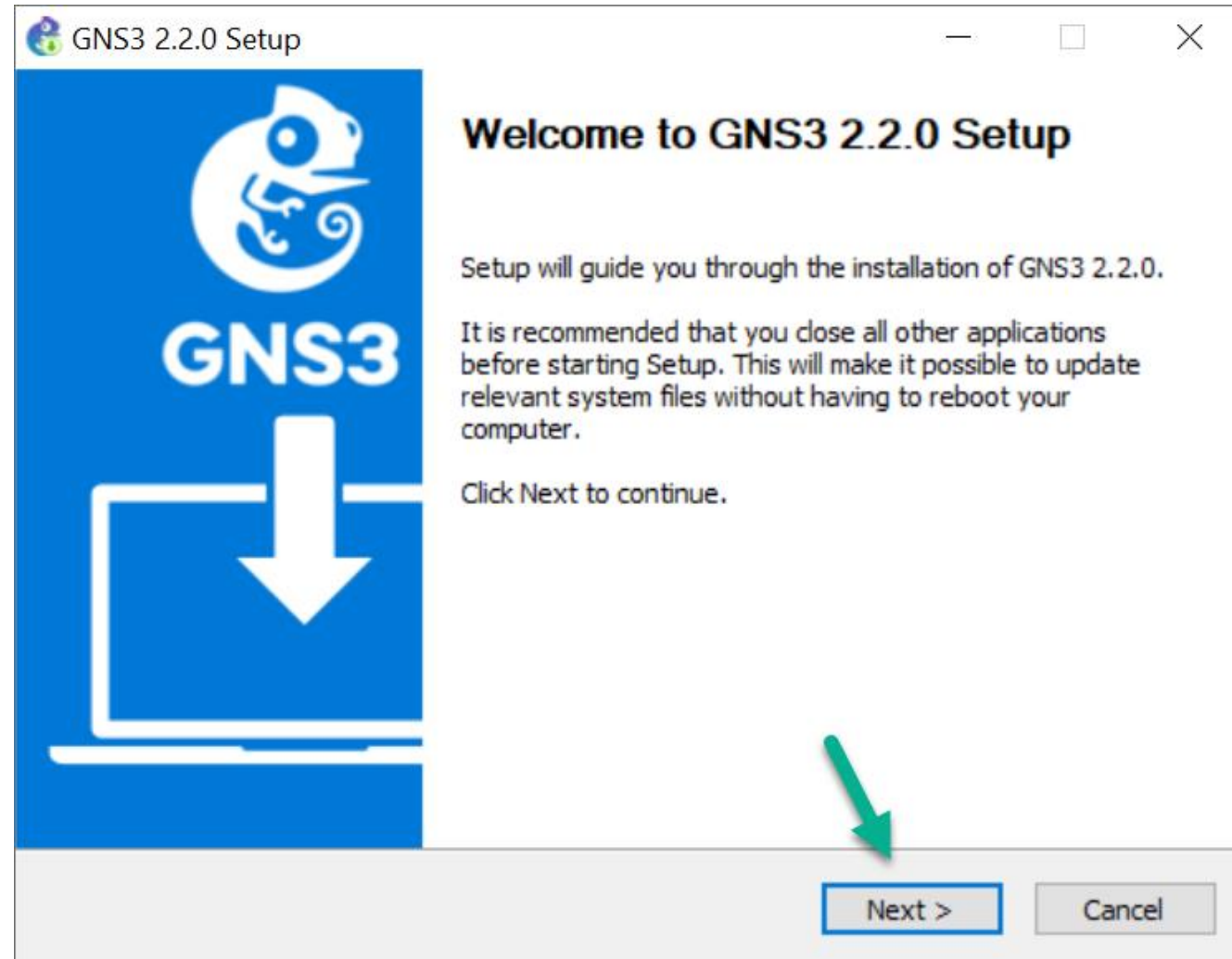
TOOLS

- ✓ GNS3
- ✓ Virtual box, Vmware
- ✓ Windows 10, Mint, ubuntu desktop, ubuntu server
(serveur apache,ssh,telnet, samba)
- ✓ Cisco IOS
- ✓ Cisco routeur et switch physique (cisco lab)

BIBLIOGRAPHIE

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GNS3 ET VMGNS3



TELNET

Routeur telnet configuration et ip add

```
en
config t
int e1/0
ip add 192.168.1.2 255.255.255.0
no shut
Exit
enable password toujours
username ebedon password toujours
username ebedon privilege 15
line vty 0 4
login local
transport input all
exit
```

Switch telnet configuration et ip add

```
en
config t
int vlan 1
ip add 192.168.1.2 255.255.255.0
no shut
exit
enable password toujours
username ebedon password toujours
line vty 0 4
login local
transport input all
Exit
ip default-gateway 192.168.1.1
```

SSH

Routeur ssh configuration et ip add

```
en
config t
host S1
int e1/0
ip add 192.168.1.2 255.255.255.0
no shut
exit
ip domain-name unh.com
crypto key generate rsa
[768]
```

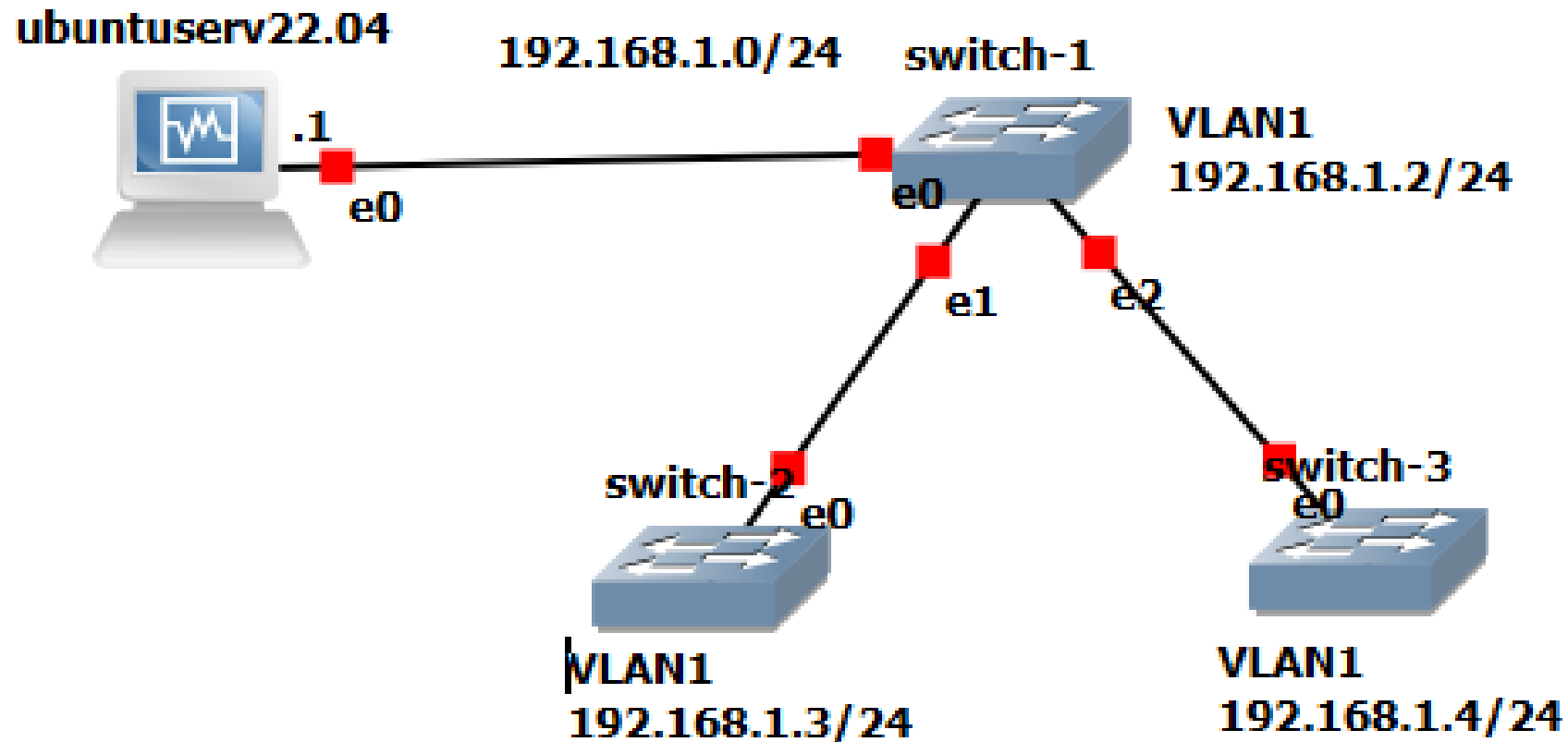
```
enable password cisco
username ebedon password cisco
username ebedon privilege 15
line vty 0 4
login local
transport input all
exit
```

Switch telnet configuration et ip add

```
en
config t
int vlan 1
ip add 192.168.1.2 255.255.255.0
no shut
exit
ip domain-name unh.com
crypto key generate rsa
[768]
```

```
enable password cisco
username ebedon password cisco
username ebedon privilege 15
line vty 0 4
login local
transport input all
Exit
ip default-gateway 192.168.1.1
```


VTP(Virtual trunking protocol), DTP (Dynamic trunking protocol)



DTP (Dynamic trunking protocol)

DTP stands for dynamic trunking protocol. DTP allows Cisco switches to dynamically learn their interface status (access or trunk) without manual configuration. It means when one switch interface is configured, other switch automatically learn the status (doesn't wait for manual configuration). For example two switches(S1 and S2) are connected together, S1 is configured as trunk port manually then S2 dynamically learn the status.

It is Cisco proprietary protocol.

By default, DTP is enabled on all Cisco switches. To disable DTP on an interface use `switchport nonegotiate` command.

	Dynamic Auto	Dynamic Desirable	Trunk	Access
Dynamic Auto	Access	Trunk	Trunk	Access
Dynamic Desirable	Trunk	Trunk	Trunk	Access
Trunk	Trunk	Trunk	Trunk	Limited Connectivity
Access	Access	Access	Limited Connectivity	Access

Note: Table assumes DTP is enabled at both ends.

Switchport mode access
 Switchport mode trunk
 Switchport mode dynamic auto
 Switchport mode dynamic desirable

VTP(Virtual trunking protocol)

“VTP allows a network manager to configure a switch so that it will propagate VLAN configurations to other switches in the network”

VTP minimizes misconfigurations and configuration inconsistencies that can cause problems, such as duplicate VLAN names or incorrect VLAN-type specifications. VTP helps you simplify management of the VLAN database across multiple switches.

VTP is a Cisco-proprietary protocol and is available on most of the Cisco switches.

I have just created VLAN 10 so I'll send VTP Update to others

I am in VTP Transparent mode so I don't care!
I just forward it to others

I am in VTP Client mode so I'll update my VLAN database!



VLAN

vlan 10
name info

interface fa0/1
switchport mode access
switchport access vlan 10

interface range fa0/1-5
switchport access vlan 10

interface fa0/1
switchport trunk encapsulation dot1q
switchport mode trunk

show vlan brief
Show int trunk

Switchport trunk allowed vlan 1-60

vtp mode client
vtp domain domain-name
vtp password test
vtp v2-mode

show vtp status
show vtp counters

CONFIGURATION